

TABLE 1 Characteristics of the included studies and overall risk of bias assessment (RoB 2) for gingivitis/mucositis.

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Study design	triple-blind, parallel-group randomized controlled trial	split-mouth clinical trial	double-blind, parallel-group randomized controlled trial	prospective, single-blind, simple randomized controlled trial	double-blind, parallel, randomized controlled trial	open-label controlled trial with two parallel arms	prospective open-label clinical trial
Year	2026	2019	2021	2021	2024	2021	2019
Country	Iran	India	India	USA	Saudi Arabia	Iraq	Iraq
Participants	36 patients with gingivitis (19 F, 17 M)	24 patients with moderate-to-severe gingivitis (25–45 years; NR)	30 participants with gingivitis (18–28 years; NR)	60 children with gingivitis (12–18 years; NR)	63 participants with gingivitis (18–40 years; NR)	52 patients with chemotherapy-induced oral mucositis (<25 to >45 years; 27 M, 25 F)	40 patients with chemotherapy-induced oral mucositis (28 M, 12 F)
Intervention	20% <i>Nigella sativa</i> mouthwash, 10 mL	<i>Nigella sativa</i> ethanolic extract (1:3, 1:1, 3:1 in glycerin), 1 mL	20% <i>Nigella sativa</i> mouthwash, 10 mL	0.1% <i>Nigella sativa</i> gel (pea-sized amount)	Standard care + 5% <i>Nigella sativa</i> gel	Commercial <i>Nigella sativa</i> oil mouthwash, 10 mL	Commercial <i>Nigella sativa</i> oil mouthwash, 10 mL
Frequency	twice daily	repeated for 3 consecutive days	twice daily	twice daily	twice daily	four times daily	five times daily
mg TQ/ Dose	NR	NR	NR	NR	NR	NR	NR
Comparator	CHX 0.2% mouthwash	SRP alone	CHX 0.12% mouthwash	CHX 0.12% gel	SC + placebo gel, SC alone	moutwash ^a	moutwash ^a
Outcome	significant reductions in PI and PBI	significant reductions in PI and GI	significant reductions in PI and GI	significant reductions in PI and GI	significant reductions in PI and PBI	significant reductions in OMAS, VAS pain, VAS swallowing and salivary IL-6	significant reductions in VRS mucositis severity and salivary IL-6 + TNFa
Follow-Up	2 weeks	28 days	2 weeks	28 days	2 weeks	28 days	60 days
Overall RoB	some concerns	some concerns	some concerns	low risk	low risk	some concerns	some concerns

Ns, *Nigella sativa*; SRP, scaling and root planing; SC, standard protocol of scaling; PI, plaque index; PBI, papillary bleeding index; GI, gingival index; OMAS, oral mucositis assessment scale; VAS, visual analogue scale, IL-6 interleukin-6, TNF- α tumor necrosis factor- α , NR not reported.

^anystatin 1,00,000 U tetracycline 0.02% lidocaine 0.5%, dexamethasone 0.5%.

In terms of study design, one of the five gingivitis studies was a triple-blind (22) and one (23) a double-blind, parallel-group randomized controlled trial. One study had a split-mouth design (24) and two studies were randomized (25, 26), one of them single-blind (25) (Table 1). All studies were exploratory and none was explicitly designed as a confirmatory efficacy trial. The two mucositis studies (27, 28) were open-label studies. Thus, considerable heterogeneity was observed in the study design, but also in the *Nigella* formulations, TQ concentrations, dosages, frequency of application and treatment duration. Four studies used a *Nigella* mouthwash, two a gel formulation that differed in their galenic and one study used an ethanolic extract in three dilutions with glycerin. The daily dose of TQ has not been reported in any of the studies. The overall risk of bias was some concerns in five of the seven studies, only two studies had a low risk of bias. Detailed domain-specific RoB 2 assessments and their justifications are presented in Supplementary Tables S4, S5. Overall, the evidence for the

effectiveness of *Nigella sativa* preparations in gingivitis and oral mucositis remains limited.

3.3 In peri-implantitis/periodontitis

The study characteristics are summarized in Table 2. Detailed domain-specific RoB 2 assessments and the rationale for the overall judgements are presented in Supplementary Tables S4, S5. One prospective randomized controlled clinical trial investigated patients suffering from peri-implantitis (29). Four of the periodontitis studies were prospective controlled clinical trials (30–33) of which only one was randomized (30). One study had a randomized single-blind split-mouth clinical trial design (34) and one was a parallel-arm randomized controlled, triple-blinded trial (35). All studies were exploratory, none was adequately powered for efficacy confirmation. Considerable heterogeneity was observed in the study design, but also in the intervention, the frequency of application, the *Nigella sativa*